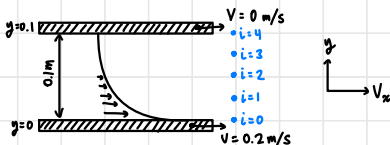


WRITTEN WORK:

TASK 1:



TASK 2:

From the 1-D flow of Newtonian Fluid Equation: $\frac{\partial V_x}{\partial t} = \frac{\mu}{\rho} \frac{\partial^2 V_x}{\partial y^2}$.

We discretize the time derivative using the Crank Nicholson Scheme:

$$\frac{\partial V_x}{\partial t} \approx \frac{V_{x,i}^{s+1} - V_{x,i}^s}{\Delta t} \quad \Delta t \text{ is the time step}$$

(i refers to node, s is current time step)

We discretize the spatial second derivative using the avg of implicit and explicit schemes

$$\frac{\partial^2 V_x}{\partial y^2} \approx \frac{1}{2} \left[\frac{V_{x,i+1}^{s+1} - 2V_{x,i}^{s+1} + V_{x,i-1}^{s+1} + V_{x,i+1}^s - 2V_{x,i}^s + V_{x,i-1}^s}{(\Delta y)^2} \right] \quad \Delta y \text{ is the space step}$$

(i refers to node, s is current time step)

Subbing into the original PDE:

$$\frac{V_{x,i}^{s+1} - V_{x,i}^s}{\Delta t} = \frac{\mu}{2\rho} \left[\frac{V_{x,i+1}^{s+1} - 2V_{x,i}^{s+1} + V_{x,i-1}^{s+1} + V_{x,i+1}^s - 2V_{x,i}^s + V_{x,i-1}^s}{(\Delta y)^2} \right]$$

$$V_{x,i}^{s+1} - V_{x,i}^s = \left(\frac{\mu \Delta t}{2\rho (\Delta y)^2} \right) (V_{x,i+1}^{s+1} - 2V_{x,i}^{s+1} + V_{x,i-1}^{s+1} + V_{x,i+1}^s - 2V_{x,i}^s + V_{x,i-1}^s)$$

Since $\lambda = \frac{\mu \Delta t}{2\rho (\Delta y)^2}$, we can substitute and expand:

$$V_{x,i}^{s+1} - V_{x,i}^s = \lambda V_{x,i+1}^{s+1} - 2\lambda V_{x,i}^{s+1} + \lambda V_{x,i-1}^{s+1} + \lambda V_{x,i+1}^s - 2\lambda V_{x,i}^s + \lambda V_{x,i-1}^s$$

$$-\lambda V_{x,i+1}^{s+1} + (1+2\lambda)V_{x,i}^{s+1} - \lambda V_{x,i-1}^{s+1} = \lambda V_{x,i+1}^s + (1-2\lambda)V_{x,i}^s + \lambda V_{x,i-1}^s$$

TASK 3:

LOWER SURFACE: $V_x(t, y=0) = 0.2 \text{ m/s} \quad [i=0] \Rightarrow V_{x,0}^{s+1} = V_{x,0}^s = 0.2 \text{ m/s}$

TOP SURFACE: $V_x(t, y=0.1) = 0 \text{ m/s} \quad [i=n+1=4] \Rightarrow V_{x,4}^{s+1} = V_{x,4}^s = 0$

$[i=1]: -\lambda V_{x,2}^{s+1} + (1+2\lambda)V_{x,1}^{s+1} - \lambda(0.2) = \lambda V_{x,2}^s + (1-2\lambda)V_{x,1}^s + \lambda(0.2)$

$[i=n=3]: -\lambda(0) + (1+2\lambda)V_{x,3}^{s+1} - \lambda V_{x,2}^{s+1} = \lambda(0) + (1-2\lambda)V_{x,3}^s + \lambda V_{x,2}^s$

TASK 4:

i	EQUATION
0	$V_{x,0}^{s+1} = V_{x,0}^s = 0.2$
1	$-\lambda V_{x,2}^{s+1} + (1+2\lambda)V_{x,1}^{s+1} = \lambda V_{x,2}^s + (1-2\lambda)V_{x,1}^s + \lambda(0.4)$
2	$-\lambda V_{x,3}^{s+1} + (1+2\lambda)V_{x,2}^{s+1} - \lambda V_{x,1}^{s+1} = \lambda V_{x,3}^s + (1-2\lambda)V_{x,2}^s + \lambda V_{x,1}^s$
3	$(1+2\lambda)V_{x,3}^{s+1} - \lambda V_{x,2}^{s+1} = (1-2\lambda)V_{x,3}^s + \lambda V_{x,2}^s$
4	$V_{x,4}^{s+1} = V_{x,4}^s = 0$

$$\Rightarrow \begin{bmatrix} (1+2\lambda) & -\lambda & 0 \\ -\lambda & (1+2\lambda) & -\lambda \\ 0 & -\lambda & (1+2\lambda) \end{bmatrix} \begin{bmatrix} V_{x,1}^{s+1} \\ V_{x,2}^{s+1} \\ V_{x,3}^{s+1} \end{bmatrix} = \begin{bmatrix} (1-2\lambda)V_{x,1}^s + \lambda V_{x,2}^s + 0.4\lambda \\ V_{x,1}^s + (1-2\lambda)V_{x,2}^s + \lambda V_{x,3}^s \\ \lambda V_{x,2}^s + (1-2\lambda)V_{x,3}^s \end{bmatrix}$$